

Unit 2 (Topic 3): All Definitions

Term	Definition
Cell theory	Concept stating all living organisms are made of cells, cells are the basic units of life, and cells arise from pre-existing cells.
Eukaryotic cell	A cell with a membrane-bound nucleus and organelles (e.g. animal, plant cells).
Prokaryotic cell	A cell without a nucleus or membrane-bound organelles (e.g. bacteria).
Ultrastructure	Detailed internal structure of a cell visible only under an electron microscope.
Organelles	Structures within a cell specialised for specific functions (e.g. mitochondria, ribosomes, Golgi).
Magnification	The degree to which an image is larger than the actual object.
Resolution	The ability to distinguish two points as separate; clarity of an image.
Mitosis	Nuclear division producing two genetically identical daughter cells.
Meiosis	Type of nuclear division producing four genetically unique haploid gametes.
Cell cycle	Series of stages (interphase and mitosis) through which a cell passes from one division to the next.
Interphase	Cell cycle phase (G1, S, G2) where the cell grows, replicates DNA, and prepares for division.
Gamete	Haploid sex cells (sperm and egg) formed by meiosis, used in sexual reproduction.
Fertilisation	Fusion of haploid gametes (egg and sperm) producing a diploid zygote.
Zygote	Diploid cell formed by fertilisation, develops into an embryo.
Stem cell	Undifferentiated cell capable of differentiating into various cell types.
Totipotent stem cell	Stem cell able to differentiate into all cell types, including placental tissue (e.g. zygote, early embryonic cells).
Pluripotent stem cell	Stem cell able to differentiate into almost all cell types (excluding placental tissue, e.g. embryonic stem cells).
Morula	Early embryonic stage consisting of a solid ball of undifferentiated totipotent cells.
Blastocyst	Hollow ball of cells containing a pluripotent inner cell mass.
Differentiation	Process by which unspecialised cells become specialised for specific functions.

Differential gene expression	Expression of specific genes leading to cell specialisation.
Epigenetics	Study of heritable changes in gene expression without alteration of DNA sequence (e.g. methylation, histone modification).
DNA methylation	Addition of methyl groups to DNA, typically suppressing gene expression.
Histone modification	Chemical changes to histone proteins affecting gene expression by altering DNA packaging.
Polygenic inheritance	Traits controlled by multiple genes, resulting in continuous variation (e.g. height, skin colour).
Continuous variation	Variation in a population showing a range of phenotypes, not discrete categories.
Phenotype	Observable physical and biochemical characteristics of an organism determined by genotype and environment.
Genotype	Genetic makeup of an organism; combination of alleles.
Mutations	Random changes in DNA sequence, potentially altering gene expression and phenotype.
Substitution mutation	Mutation replacing one nucleotide with another; may or may not alter amino acid sequences.
Deletion mutation	Mutation removing nucleotides, causing frameshift; affects many amino acids downstream.
Insertion mutation	Mutation adding nucleotides, causing frameshift; affects many amino acids downstream.
Tumour	Mass of abnormal cells resulting from uncontrolled cell division.
Cancer	Malignant tumour that invades tissues and can spread (metastasise).
Benign tumour	Non-cancerous tumour that does not invade other tissues or metastasise.
Malignant tumour	Cancerous tumour capable of invading surrounding tissue and metastasising.
Oncogene	Mutated proto-oncogene causing uncontrolled cell division, contributing to cancer development.
Tumour suppressor gene	Gene inhibiting cell division and promoting apoptosis; mutations contribute to cancer development when inactivated.
Metastasis	Spread of cancer cells from original site to other parts of the body via blood or lymph.

Crossing over	Exchange of genetic material between homologous chromosomes during meiosis, increasing genetic variation.
Independent assortment	Random orientation of homologous chromosome pairs during meiosis I, contributing to genetic variation.
Acrosome reaction	Release of enzymes from sperm acrosome to penetrate egg's zona pellucida during fertilisation.
Cortical reaction	Release of enzymes from cortical granules in egg after fertilisation, preventing polyspermy.
Zona pellucida	Glycoprotein layer surrounding egg cells, involved in preventing polyspermy.
Haploid cell (n)	Cell containing a single set of chromosomes (gametes).
Diploid cell (2n)	Cell containing two sets of chromosomes, one from each parent.

Unit 2 Topic 4: All Definitions

Term	Definition
Biodiversity	The variety of living organisms in an area, encompassing species, genetic, and habitat diversity.
Endemism	The ecological state of being unique to a defined geographic location and found nowhere else.
Species	A group of organisms capable of interbreeding and producing fertile offspring.
Habitat	The specific environment where an organism lives, characterized by physical conditions and other species present.
Species richness	The total number of different species present within a habitat or ecosystem.
Genetic diversity	The number and frequency of alleles present within the gene pool of a species or population.
Niche	The role or function of a species within its habitat, including its interactions and use of resources.
Adaptation	An inherited feature (anatomical, behavioural, physiological) that increases an organism's chance of survival.
Natural selection	The process by which individuals with advantageous traits survive and reproduce more, passing on these traits.
Evolution	A change in allele frequencies within a population over generations.
Speciation	The formation of new and distinct species through evolutionary processes.
Allopatric speciation	Speciation resulting from geographic isolation preventing gene flow between populations.
Sympatric speciation	Speciation occurring within the same geographic area due to reproductive isolation.
Reproductive isolation	Mechanisms (behavioural, ecological, mechanical, temporal, genetic) preventing two species from interbreeding.
Population bottleneck	An event causing a significant reduction in a population's size, greatly reducing genetic diversity.
Genetic drift	Random changes in allele frequency within a population, often significant in small populations.
Gene flow	The transfer of genetic material (alleles) between populations through migration and reproduction.

Inbreeding depression	Reduced biological fitness resulting from mating between closely related individuals, increasing harmful alleles.
Heterozygosity index	A measure of genetic diversity calculated as the proportion of heterozygotes in a population.
Simpson's Diversity Index	A numerical measure of biodiversity accounting for species richness and evenness in a habitat.
Classification	Organising living organisms into hierarchical groups based on shared characteristics and evolutionary relationships.
Three-domain system	A classification system dividing life into three groups: Bacteria, Archaea, and Eukarya.
Cellulose	Polysaccharide consisting of β -glucose monomers; primary component of plant cell walls.
Starch	A polysaccharide energy storage molecule in plants, consisting of amylose and amylopectin.
Lignin	A rigid polymer in plant cell walls, providing structural support and waterproofing.
Xylem	Plant tissue transporting water and mineral ions; composed of dead, lignified cells.
Phloem	Plant tissue transporting organic solutes (sucrose); made of living sieve tubes and companion cells.
Sclerenchyma	Supportive plant tissue composed of dead, lignified cells without transport function.
Sustainability	Meeting present needs without compromising the ability of future generations to meet theirs.
Micropropagation	Cloning plants using tissue culture techniques to produce genetically identical individuals.
Cryopreservation	Preservation of biological material at very low temperatures (-196°C using liquid nitrogen).
Seed bank	Facility for storing seeds under controlled conditions to preserve genetic diversity and viability.
Artificial pollination	Manual transfer of pollen to ensure controlled fertilisation and genetic diversity.
Zoos	Facilities housing animals in controlled environments for conservation, breeding, and public education.

Botanical garden	Controlled environment for cultivating diverse plant species for conservation, research, and education.
Arboretum	Botanical collection specialising in the cultivation and conservation of trees.
In situ conservation	Conservation of species in their natural habitats.
Ex situ conservation	Conservation of species outside their natural habitats (e.g. zoos, seed banks, botanical gardens).
Stud book	Detailed records tracking lineage, genetic traits, and breeding history of captive organisms.
Artificial insemination	Direct introduction of sperm into a female's reproductive system without natural mating.
In vitro fertilisation	Fertilisation of an egg by sperm in laboratory conditions; embryos are implanted into females.
Gene bank	Repository for storing genetic material (sperm, eggs, embryos, seeds) to conserve genetic diversity.
Reintroduction	Release of captive-bred individuals into their natural habitats to re-establish populations.